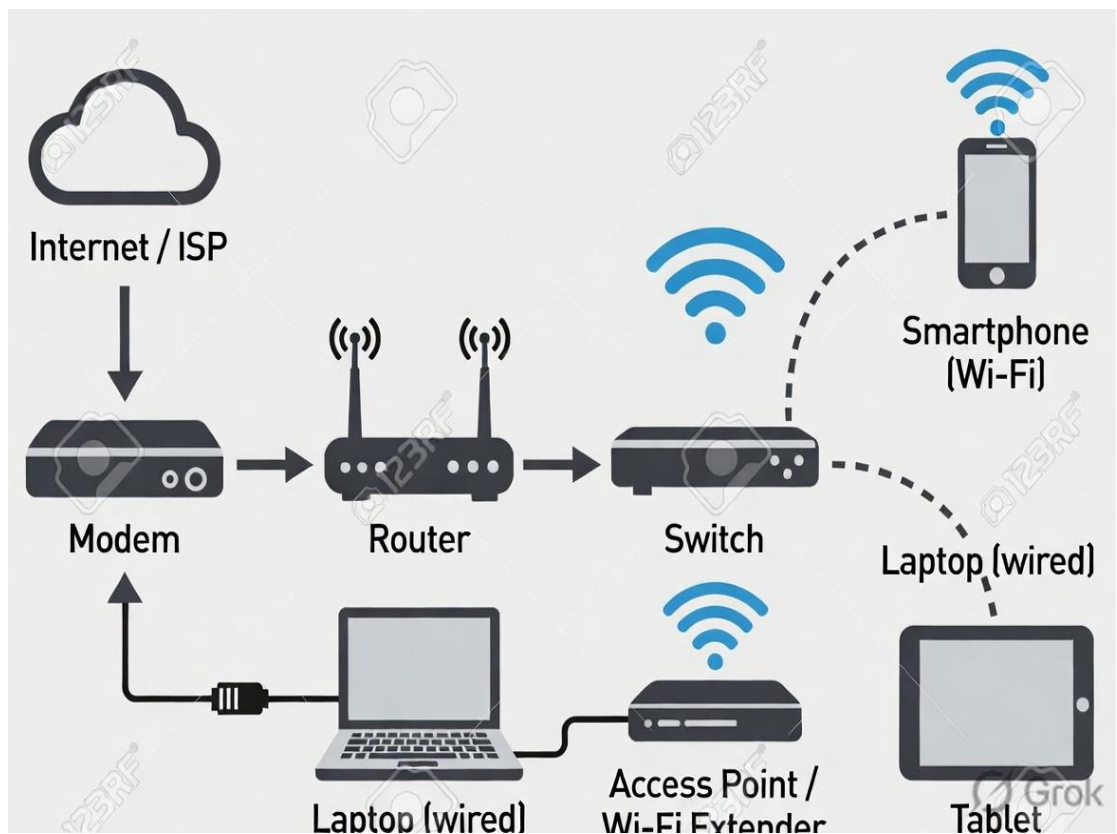


DAY-1

INTRODUCTION TO NETWORKING HARDWARE THEORY

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Modem

- . *Function: Converts analog signals from your service provider into digital data that your computers can read.*
- . *Example: Sitting next to the TV box where the main coaxial or fiber line comes in from the street.*

Router

- . *Function: Connects multiple networks together and directs data packets to their correct destinations.*
- . *Example: At home to share one internet connection among phones, laptops, and game consoles.*

Switch

- . *Function: Connects wired devices inside a single local network and passes data only to the intended target.*
- . *Example: In a small office desk setup to plug four desktop PCs into the same local network using cords.*

Access Point

- . *Function: Creates a wireless Wi-Fi signal to let devices join the network without wires.*
- . *Example: Mounted on the ceiling of a large college library so students can connect laptops to the internet from anywhere in the room.*

Network Cable

- . *Function: Carries physical data signals safely between computers, routers, and switches.*

- *Example: An Ethernet cord plugged directly from a smart TV into the back of a living room router for a stable streaming connection.*
3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet



- *Twisted-pair Ethernet cable (Cat5e/Cat6): standard for LANs and most office/home networking;*

2. Coaxial



Coaxial cable (coax): commonly used for older cable TV/internet and many broadband connections .

3. Fiber optic



Fiber optic cable (single-mode/multimode): used for high-speed backbone, long-distance links, and data centers.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

SWITCHING VS ROUTING		
FEATURE	SWITCHING	ROUTING
Purpose	Forwards frames within the same LAN, creates connectivity inside a local network.	Forwards packets between different networks (LAN ↔ WAN ↔ Internet).
Device & Layer	Uses Switch works at Data Link Layer (Layer 2).	Uses Router / L3 Switch works at Network Layer (Layer 3).
Address & Data	Uses MAC address (Physical) for forwarding data unit = Frame.	Uses IP address (Logical) for forwarding data unit = Packet.
Speed & Behavior	Faster (less processing), allows broadcasts unless VLANs used.	Slower (path calculation, routing tables), does not forward broadcasts.